

More Gaussian, then Statistics

Dr Lily Asquith (Lily, she/her)



More Gaussian, then Stats

- More about the Gaussian
- Multivariate Gaussian
- The Law(s) of Large Numbers
- The Central Limit Theorem
- The (Null) Hypothesis & Confusion Matrix
- Significance, p-values, Confidence

Intro, reminder of big picture

We will spend a bit of time on the Gaussian because it is central to data science/ machine learning.

Then we will look at the two main stats laws: The (weak) Law of Large Numbers (LLN) and The Central Limit Theorem.

Then we will refresh our understanding of popular stats terms such as the Null Hypothesis, true and false positives, significance (z-values and p-values), and confidence levels.

The Gaussian Distribution

The Gaussian (aka Normal)

PDF:

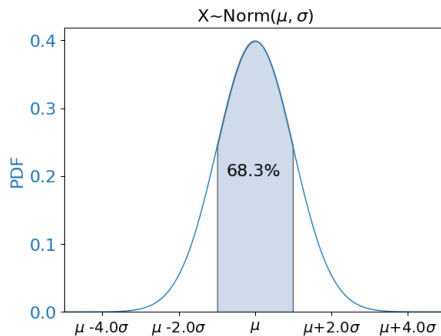
$$f_X(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$

Standardised form:

$$f_Z(x - \mu; \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2} z^2}$$

Most basic form :

$$f_Y(y) = e^{-y^2}$$

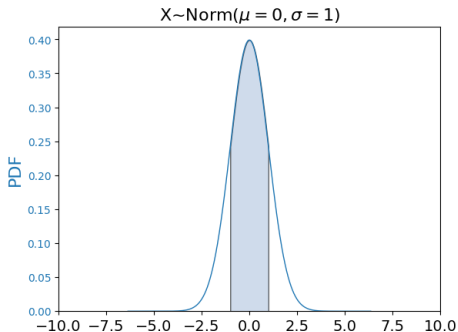


This term: $z = \frac{x-\mu}{\sigma}$ has many names: score, z-score, z-statistic, pull...

Features of the Gaussian

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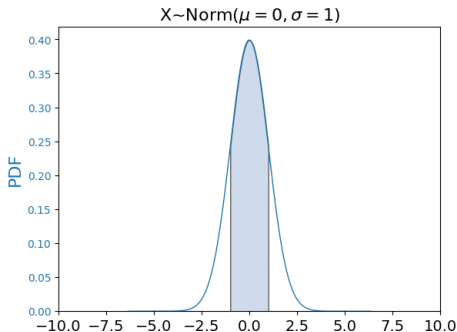
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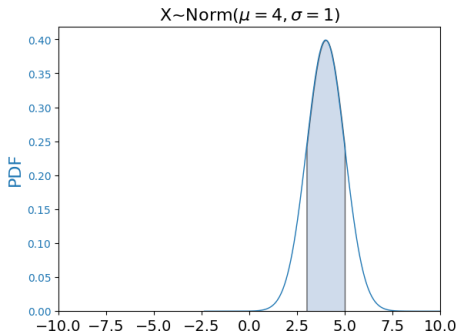
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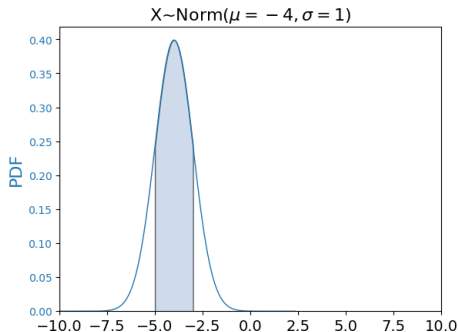
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- 3 The location parameter is μ : shift left/right



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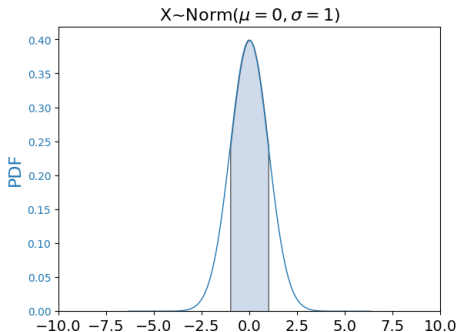
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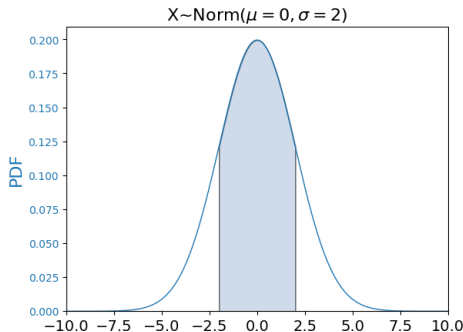
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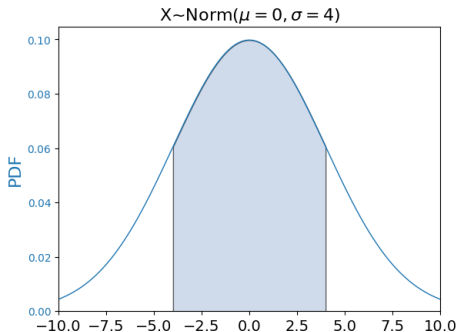
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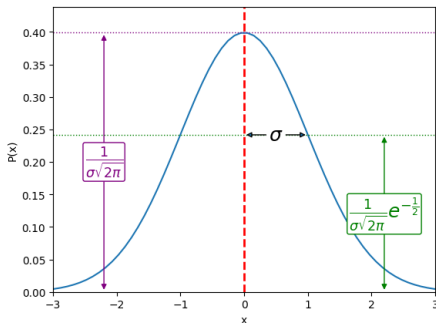
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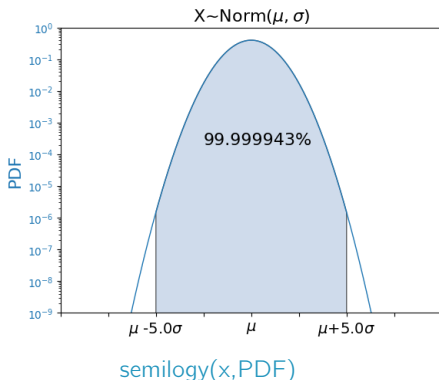
- 1 The maximum of the PDF is at $x = \mu$, because $e^0 = 1$
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More features of the Gaussian

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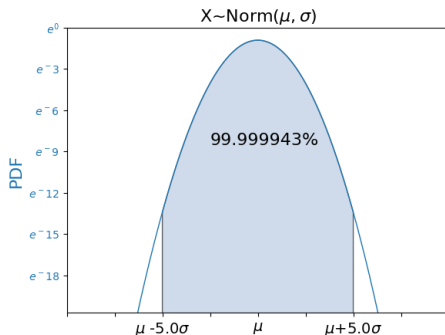
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- 2 The maximum of the PDF is $\frac{1}{\sigma\sqrt{2\pi}}$
- 3 The (natural) log (ln) of the PDF is quadratic in x .



More features of the Gaussian

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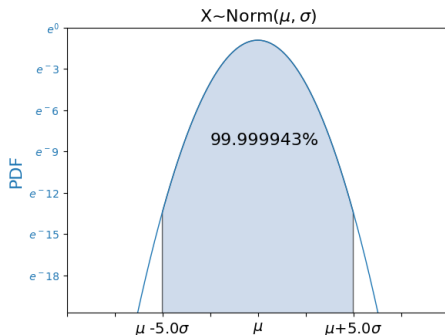


`semilogy(x,PDF, base=np.e)`

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- 3 The (natural) log (ln) of the PDF is quadratic in x .
- 4 Quadratic in x : bowl shaped
- 5 Negative quadratic: inverted bowl



`semilogy(x,PDF, base=np.e)`

Log of Gaussian PDF

$$f_X(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$

$$\ln f_X(x; \mu, \sigma) = \ln \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}} \right]$$

Log of Gaussian PDF

$$f_X(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$

$$\begin{aligned}\ln f_X(x; \mu, \sigma) &= \ln \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}} \right] \\ &= \ln \left[\frac{1}{\sigma\sqrt{2\pi}} \right] + \ln \left[e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}} \right]\end{aligned}$$

because $\ln ab = \ln a + \ln b$

Log of Gaussian PDF

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because $\ln e^a = a$

Log of Gaussian PDF

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because $\ln ab = \ln a + \ln b$

because $\ln e^a = a$

because $\ln \frac{1}{a} = -\ln a$

Log of Gaussian PDF

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because $\ln ab = \ln a + \ln b$

because $\ln e^a = a$

because $\ln \frac{1}{a} = -\ln a$

because $-\ln ab = -\ln a - \ln b$

This will be useful to us when we come to maximising the log likelihood for parameter estimation.

Gaussian: Univariate to Multivariate

Univariate:

$$f_X(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2} \frac{(x - \mu)^2}{\sigma^2}\right]$$

Multivariate:

$$f_X(x; \mu, \Sigma) = \frac{1}{|\Sigma|^{1/2} (2\pi)^{Z/2}} \exp\left[-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)\right]$$

Yikes.

Gaussian: Univariate to Multivariate

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In the normalisation we have $\sigma \rightarrow |\Sigma|^{1/2}$

We also have $\sqrt{2\pi} \rightarrow (2\pi)^{Z/2}$ to keep the PDF valid (Z is the number of RVs).

Example: Two Independent RVs [1. Determinant]

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \quad \Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \leftarrow \text{Independent RVs: } \Sigma \text{ diagonal.}$$

$$f_X(x; \mu, \Sigma) = \frac{1}{|\Sigma|^{1/2} (2\pi)^{Z/2}} \exp\left[-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right]$$

$$|\Sigma|^{1/2} = \begin{vmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{vmatrix}^{1/2} = (\sigma_1^2 \sigma_2^2 - 00)^{1/2} = \sigma_1 \sigma_2$$

Example: Two Independent RVs [2. Normalisation]

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \quad \Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \leftarrow \text{Independent RVs: } \Sigma \text{ diagonal.}$$

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$$\frac{1}{|\Sigma|^{1/2} (2\pi)^{Z/2}} = \frac{1}{\sigma_1 \sigma_2 2\pi}$$

Example: Two Independent RVs [3. Cov Matrix]

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \quad \Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

$$f_X(x; \mu, \Sigma) = \frac{1}{\sigma_1 \sigma_2 2\pi} \exp\left[-\frac{1}{2}(x - \mu)^T \Sigma^{-1} (x - \mu)\right]$$

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$$\begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{\sigma_1^2} & 0 \\ 0 & \frac{1}{\sigma_2^2} \end{bmatrix}$$

Warning! We can only invert a matrix like this if it is diagonal!

Finding the inverse of a matrix is usually a total faff.

Example: Two Independent RVs [4. Multiply]

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \quad \Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

$$f_X(x; \mu, \Sigma) = \frac{1}{\sigma_1 \sigma_2 2\pi} \exp\left[-\frac{1}{2}(x - \mu)^T \Sigma^{-1} (x - \mu)\right]$$

$$\begin{bmatrix} \frac{1}{\sigma_1^2} & 0 \\ 0 & \frac{1}{\sigma_2^2} \end{bmatrix} \begin{bmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sigma_1^2} (x_1 - \mu_1) \\ \frac{1}{\sigma_2^2} (x_2 - \mu_2) \end{bmatrix}$$

Example: Two Independent RVs [5. Multiply]

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \quad \Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

$$f_X(x; \mu, \Sigma) = \frac{1}{\sigma_1 \sigma_2 2\pi} \exp\left[-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right]$$

$$\begin{bmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \end{bmatrix}^\top \begin{bmatrix} \frac{1}{\sigma_1^2}(x_1 - \mu_1) \\ \frac{1}{\sigma_2^2}(x_2 - \mu_2) \end{bmatrix} = \frac{(x_1 - \mu_1)^2}{\sigma_1^2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2}$$

Example: $Z = 2$ independent RVs [6. Full Argument]

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \quad \Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

$$f_X(x; \mu, \Sigma) = \frac{1}{\sigma_1 \sigma_2 2\pi} \exp \left[-\frac{1}{2} \left[\frac{(x_1 - \mu_1)^2}{\sigma_1^2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2} \right] \right]$$

Example: $Z = 2$ independent RVs [6. Full Argument]

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$$e^{-A-B} = e^{-A} e^{-B}$$

$$f_X(x; \mu, \Sigma) = \frac{1}{\sigma_1 \sigma_2 2\pi} \exp \left[-\frac{(x_1 - \mu_1)^2}{2\sigma_1^2} \right] \exp \left[-\frac{(x_2 - \mu_2)^2}{2\sigma_2^2} \right]$$

Example: $Z = 2$ independent RVs [7. Result]

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \quad \Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

$$f_X(x; \mu, \Sigma) = \frac{1}{\sigma_1 \sigma_2 2\pi} \exp\left[-\frac{(x_1 - \mu_1)^2}{2\sigma_1^2}\right] \exp\left[-\frac{(x_2 - \mu_2)^2}{2\sigma_2^2}\right]$$

Example: $Z = 2$ independent RVs [7. Result]

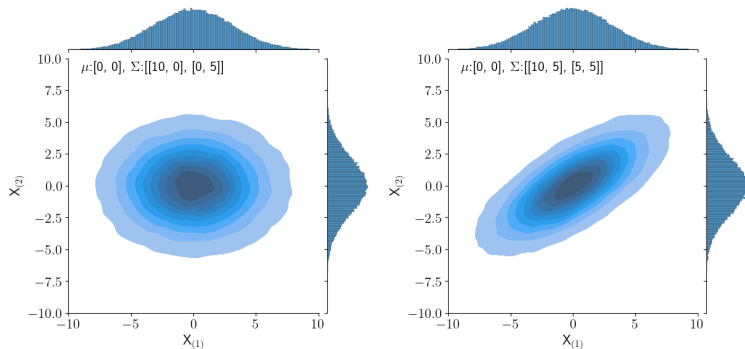
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$$f_X(x; \mu, \Sigma) = \frac{1}{\sigma_1 \sqrt{2\pi}} \exp\left[-\frac{(x_1 - \mu_1)^2}{2\sigma_1^2}\right] \frac{1}{\sigma_2 \sqrt{2\pi}} \exp\left[-\frac{(x_2 - \mu_2)^2}{2\sigma_2^2}\right]$$

A multivariate Gaussian PDF with two independent RVS is the product of the Gaussian PDF for each of the RVs.

$$f_X(x; \mu, \Sigma) = f_X(x_1; \mu_1, \sigma_1) f_X(x_2; \mu_2, \sigma_2)$$

Bivariate Gaussian Visually: Error Ellipses



On these plots the value of the PDF is mapped to colour. **Isocontours** have the same colour: same probability.

Bivariate Gaussian: Error Ellipses

Isocontours have constant probability $f_X(x; \mu, \Sigma) = c$

$$c = \frac{1}{\sigma_1 \sigma_2 2\pi} \exp \left[- \left[\frac{(x_1 - \mu_1)^2}{2\sigma_1^2} + \frac{(x_2 - \mu_2)^2}{2\sigma_2^2} \right] \right] \quad \text{Joint probability} = \text{constant}$$

Bivariate Gaussian: Error Ellipses

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$$\ln(\sigma_1 \sigma_2 2\pi c) = - \left[\frac{(x_1 - \mu_1)^2}{2\sigma_1^2} + \frac{(x_2 - \mu_2)^2}{2\sigma_2^2} \right] \quad \text{Log of both sides}$$

Bivariate Gaussian: Error Ellipses

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$$\ln \frac{1}{(\sigma_1 \sigma_2 2\pi c)} = \frac{(x_1 - \mu_1)^2}{2\sigma_1^2} + \frac{(x_2 - \mu_2)^2}{2\sigma_2^2} \quad -\ln(a) = \ln \frac{1}{a}$$

Bivariate Gaussian: Error Ellipses

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$$1 = \frac{(x_1 - \mu_1)^2}{2\sigma_1^2 (\ln \frac{1}{(\sigma_1 \sigma_2 2\pi c)})} + \frac{(x_2 - \mu_2)^2}{2\sigma_2^2 (\ln \frac{1}{(\sigma_1 \sigma_2 2\pi c)})}$$

Bivariate Gaussian: Error Ellipses

Isocontours have constant probability $f_X(x; \mu, \Sigma) = c$

$$c = \frac{1}{\sigma_1 \sigma_2 2\pi} \exp \left[- \left[\frac{(x_1 - \mu_1)^2}{2\sigma_1^2} + \frac{(x_2 - \mu_2)^2}{2\sigma_2^2} \right] \right] \quad \text{Joint probability} = \text{constant}$$

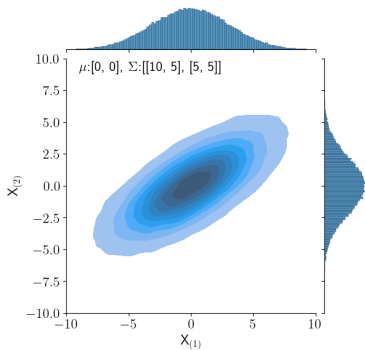
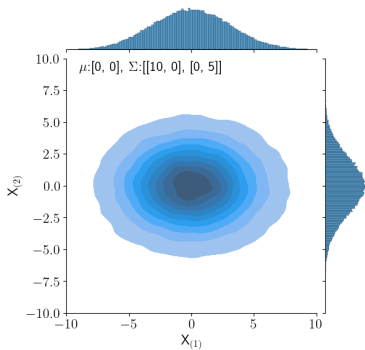
$$\ln(\sigma_1 \sigma_2 2\pi c) = - \left[\frac{(x_1 - \mu_1)^2}{2\sigma_1^2} + \frac{(x_2 - \mu_2)^2}{2\sigma_2^2} \right] \quad \text{Log of both sides}$$

$$\ln \frac{1}{(\sigma_1 \sigma_2 2\pi c)} = \frac{(x_1 - \mu_1)^2}{2\sigma_1^2} + \frac{(x_2 - \mu_2)^2}{2\sigma_2^2} \quad -\ln(a) = \ln \frac{1}{a}$$

$$1 = \frac{(x_1 - \mu_1)^2}{2\sigma_1^2 (\ln \frac{1}{(\sigma_1 \sigma_2 2\pi c)})} + \frac{(x_2 - \mu_2)^2}{2\sigma_2^2 (\ln \frac{1}{(\sigma_1 \sigma_2 2\pi c)})} \quad = \frac{(x_1 - \mu_1)^2}{r_1^2} + \frac{(x_2 - \mu_2)^2}{r_2^2}$$

This is the equation of an ellipse with semi-major axis r_1 and semi-minor axis r_2

Ellipse



Principal Components

$$1 = \frac{(x_1 - \mu_1)^2}{2\sigma_1^2 \left(\ln \frac{1}{(\sigma_1 \sigma_2 2\pi c)} \right)} + \frac{(x_2 - \mu_2)^2}{2\sigma_2^2 \left(\ln \frac{1}{(\sigma_1 \sigma_2 2\pi c)} \right)} = \frac{(x_1 - \mu_1)^2}{r_1^2} + \frac{(x_2 - \mu_2)^2}{r_2^2}$$

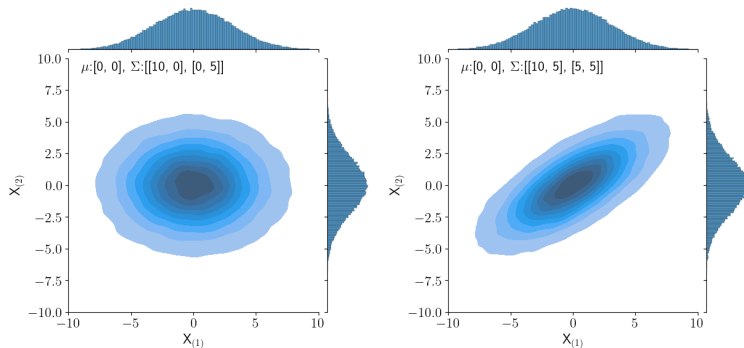
This is the equation of an ellipse with **semi-major axis** r_1 and **semi-minor axis** r_2

This is cool because the height of the Gaussian at 1σ half-width is $c_{1\sigma} = \frac{1}{e\sigma_1\sigma_2 2\pi}$ and if we plug this into our ellipse radii we get the **simple form**:

$$r_j = \sqrt{2\sigma_j^2 \left(\ln \frac{1}{\sigma_1 \sigma_j 2\pi \left(\frac{1}{e\sigma_1 \sigma_2 2\pi} \right)} \right)} = \sqrt{2}\sigma_j$$

The r_j are the **Principal Components** of the data. In this example there are only two dimensions, but we have rotated them such that they are in the directions of maximum variance.

Principal Components



Principal Component Analysis is useful when we have high dimensional data and want to reduce it to two dimensions: the two with maximum variance. Nice intro [here](#).

The Law(s) of Large Numbers

The **Weak Law of Large Numbers** (LLN):

$$\lim_{N \rightarrow \infty} P(|\bar{x} - \mu| \geq \alpha) = 0 \text{ for } \alpha > 0$$

The Weak LLN states that there is a high probability that the sample mean $\bar{x}$ will approach the expected value (true mean $E[X] \equiv \mu$) in the limit of an infinite number of measurements.

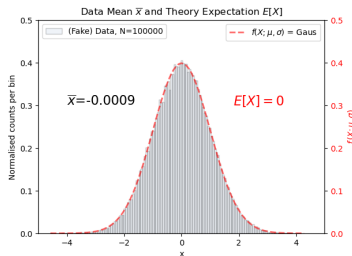
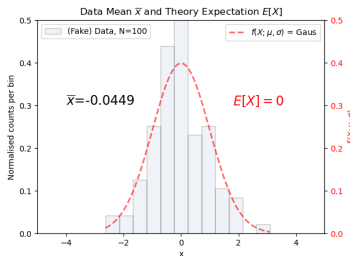
The Strong LLN: $P(\lim_{N \rightarrow \infty} \bar{x} = \mu) = 1$.

The difference between these is fairly subtle, with the Strong LLN indicating **almost sure** convergence (almost sure is a statistical term).

The Weak LLN

$$\lim_{N \rightarrow \infty} P(|\bar{x} - \mu| \geq \alpha) = 0 \text{ for } \alpha > 0$$

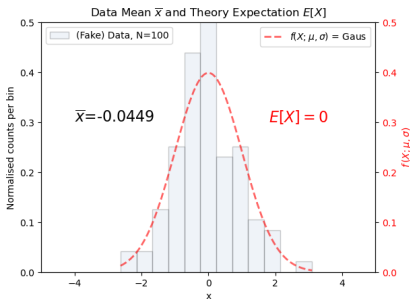
We have referred to the Weak LLN previously when we looked at how the mean tends to approach the expected value as we increase N :



The $N = 100$ experiment has $|\bar{x} - \mu| = 0.0449 = \alpha_1$. If we have chosen a tolerance of $\alpha = 0.1$, then $\alpha_1 < \alpha$: a successful outcome. If we have chosen a tolerance of $\alpha = 10^{-9}$, then $\alpha_1 > \alpha$: this is a failure.

The Weak LLN

$$\lim_{N \rightarrow \infty} P(|\bar{x} - \mu| \geq \alpha) = 0 \text{ for } \alpha > 0$$



← For this **single experiment**

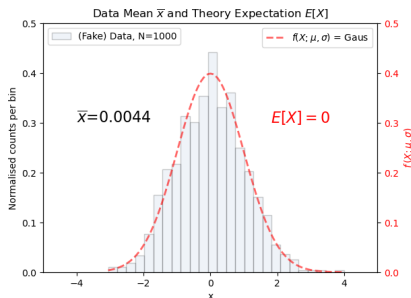
$$|\bar{x} - \mu| = 0.0449 = \alpha_1$$

If we were to repeat the $N = 100$ experiment in the plot a thousand times ($Z=1k$), we would have a thousand α_j values. Some fraction of these will fall within our **tolerance**.

The plot above yields a single $\alpha_j = \alpha_1$: it is just one of our one thousand experiments.

The Weak LLN

$$\lim_{N \rightarrow \infty} P(|\bar{x} - \mu| \geq \alpha) = 0 \text{ for } \alpha > 0$$



← For this **single experiment**

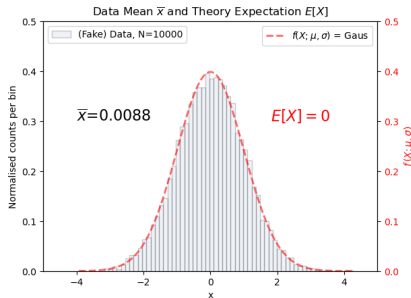
$$|\bar{x} - \mu| = 0.0044 = \alpha_1$$

If we increase $N = 1000$, and keep $Z=1k$, we will find that a larger fraction of α_j fall within our **tolerance**.

Again, the plot above yields a single α_j : it is just one of our $Z=1k$ experiments.

The Weak LLN

$$\lim_{N \rightarrow \infty} P(|\bar{x} - \mu| \geq \alpha) = 0 \text{ for } \alpha > 0$$



← For this **single experiment**

$$|\bar{x} - \mu| = 0.0088 = \alpha_1$$

If we increase $N = 10,000$, we will find that an even larger fraction of α_j fall within our **tolerance**.

The α_1 for this single experiment has increased! That's okay, it is just one of our $Z=1k$ experiments.

The Weak LLN

$$\lim_{N \rightarrow \infty} P(|\bar{x} - \mu| \geq \alpha) = 0 \text{ for } \alpha > 0$$

In the limit where our sample size N approaches ∞ , the fraction of means falling outside our tolerance approaches zero. This is true for any non-zero value of tolerance, α .

This is a Probability Limit. Can also be written: $\text{plim}_{N \rightarrow \infty} \bar{x} = \mu$

This does not mean that the sample mean approaches the population mean as the sample size grows. We have seen that is not necessarily the case.

The Weak LLN tells us that the probability converges. It does not tell us that the value of the sample mean converges.

The Weak LLN only applies to the mean, not to the variance or other statistics.

Bernoulli's Weak LLN

$$\lim_{N \rightarrow \infty} P(|\bar{x} - \mu| \geq \alpha) = 0 \text{ for } \alpha > 0$$

Bernoulli's LLN:

$$\lim_{N \rightarrow \infty} P\left(\left|\frac{k}{n} - p\right| \geq \alpha\right) = 0 \text{ for } \alpha > 0$$

Here $\frac{k}{n}$ is the (measured) frequency of success and p is the (theoretical) probability of a successful outcome.

Recall that a Bernoulli trial repeated n times is Binomial. The RV is K and its expected value is np . So this is not any different from the regular weak LLN.

This form is particularly useful for comparing histograms of data with PDFs, as the number of entries in a density histogram bin is a frequency for a range of the RV, and the area under the PDF is a probability for a range of the RV.

Expected Value of the mean ('Mean of means')

In real life we will not have infinite datasets, but finite samples. We will calculate the sample mean $\bar{x}$. The Expected Value of the sample mean is the population mean.

$$E[\bar{x}] = E\left[\frac{1}{N} \sum_i^N x_i\right] \text{ because } \bar{x} = \frac{1}{N} \sum_i^N x_i$$

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The expected value of the sample mean of the independent, identically distributed (iid) measurements x_i is the population (true) mean. Or, **the mean of the sample means is the population (true) mean.**

Variance on the mean

$$V[\bar{x}] = V\left[\frac{1}{N} \sum_i^N x_i\right] \text{ because } \bar{x} = \frac{1}{N} \sum_i^N x_i$$

Variance on the mean

$$\begin{aligned} V[\bar{x}] &= V\left[\frac{1}{N} \sum_i^N x_i\right] \text{ because } \bar{x} = \frac{1}{N} \sum_i^N x_i \\ &= \sum_i^N V\left[\frac{1}{N} x_i\right] \text{ because } V[\text{sum}] = \text{sum}(V) \text{ for iid } x_i \end{aligned}$$

Variance on the mean

$$\begin{aligned}V[\bar{x}] &= V\left[\frac{1}{N} \sum_i^N x_i\right] \text{ because } \bar{x} = \frac{1}{N} \sum_i^N x_i \\&= \sum_i^N V\left[\frac{1}{N} x_i\right] \text{ because } V[\text{sum}] = \text{sum}(V) \text{ for iid } x_i \\&= \frac{1}{N^2} \sum_i^N V[x_i] \text{ because } V[ax] = a^2 V[x]\end{aligned}$$

Variance on the mean

$$\begin{aligned} V[\bar{x}] &= V\left[\frac{1}{N} \sum_i^N x_i\right] \text{ because } \bar{x} = \frac{1}{N} \sum_i^N x_i \\ &= \sum_i^N V\left[\frac{1}{N} x_i\right] \text{ because } V[\text{sum}] = \text{sum}(V) \text{ for iid } x_i \\ &= \frac{1}{N^2} \sum_i^N V[x_i] \text{ because } V[ax] = a^2 V[x] \\ &= \frac{1}{N^2} NV[X] \text{ because } V[X] = \frac{1}{N} \sum_i^N (x_i - \bar{x})^2 = \frac{1}{N} \sum_i^N V[x_i] \end{aligned}$$

Variance on the mean

$$\begin{aligned} V[\bar{x}] &= V\left[\frac{1}{N} \sum_i^N x_i\right] \text{ because } \bar{x} = \frac{1}{N} \sum_i^N x_i \\ &= \sum_i^N V\left[\frac{1}{N} x_i\right] \text{ because } V[\text{sum}] = \text{sum}(V) \text{ for iid } x_i \\ &= \frac{1}{N^2} \sum_i^N V[x_i] \text{ because } V[ax] = a^2 V[x] \\ &= \frac{1}{N^2} NV[X] \text{ because } V[X] = \frac{1}{N} \sum_i^N (x_i - \bar{x})^2 = \frac{1}{N} \sum_i^N V[x_i] \\ &= \frac{1}{N} V[X] \end{aligned}$$

The variance of the sample mean of the iid measurements is $\frac{1}{N}$ times the population (true) variance, $V[X]$.

Variance on the mean (SEM: $\sigma_{\bar{x}}$)

$$V[\bar{x}] = \frac{1}{N} V[X]$$

This is an important result because it tells us that the uncertainty on the sample mean, $\sigma_{\bar{x}}$, goes like $\frac{1}{\sqrt{N}}$, where N is the number of measurements.

$$\sigma_{\bar{x}} = \frac{1}{\sqrt{N}} \sigma_{true}$$

$\sigma_{\bar{x}}$ is known as the **Standard Error on the Mean** (SEM).

- σ_{true} is irreducible: it is the true spread of values of a RV, because the RV is R:Random!
- The measurement uncertainty $\sigma_{\bar{x}}$ will become smaller as we take more data, and has the true spread of the RV as a constant factor.

The Central Limit Theorem

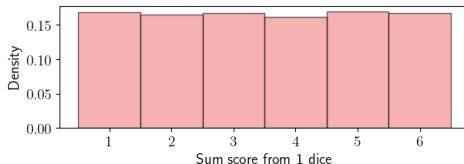
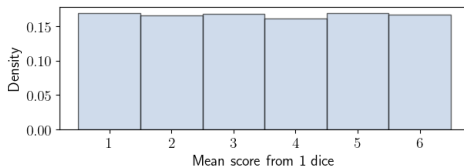
The Central Limit Theorem (CLT) states that the **distribution of sample means** $\bar{x}$ from any underlying PDF converges to a Gaussian distribution with mean μ and variance σ^2/N , as N approaches infinity. The N here is the number of measurements used to calculate the means.

The mean is a normalised sum of N measurements, so the CLT equivalently states that the **distribution of the sum of N RVs** converges to Gaussian.

The means/RVs do not have to come from the Gaussian PDF - they can come from any PDF. If you calculate $N \rightarrow \infty$ means (or sums) of RVs from a uniform, binomial, exponential or any finite variance PDF, the means will be normally distributed.

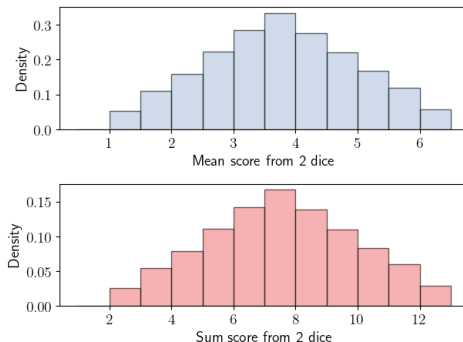
The Central Limit Theorem, Visually

N=1: I am calculating the mean (sum) of a single measurement (or a single RV).
(Yes, this is silly, because the mean of one value is just the value!) To fill the histogram, I have repeated the 1-measurement experiment Z times.



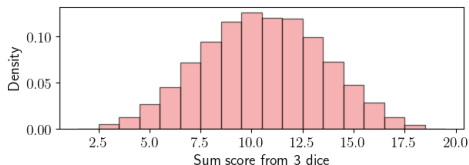
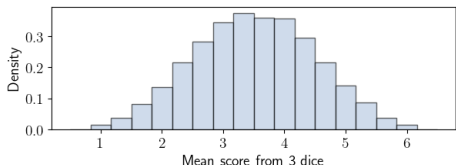
The Central Limit Theorem, Visually

$N=2$: I am calculating the mean (sum) of two measurements (or, two RVs).



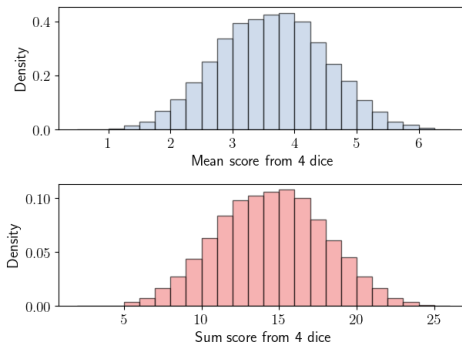
The Central Limit Theorem, Visually

N=3: I am calculating the mean (sum) of three measurements (or, three RVs).



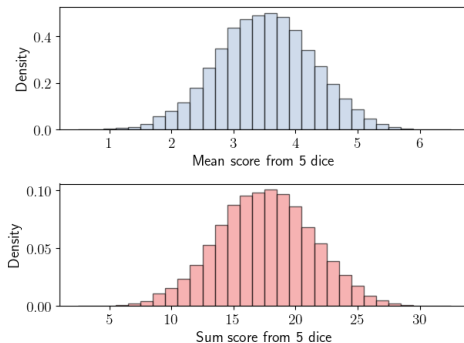
The Central Limit Theorem, Visually

And so on.



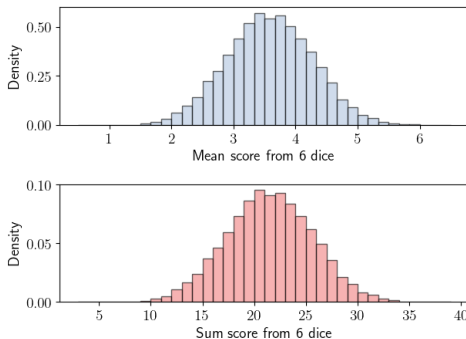
The Central Limit Theorem, Visually

And so on.



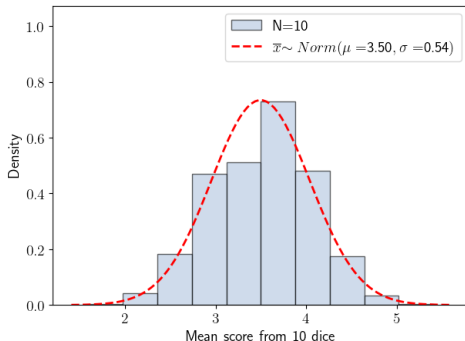
The Central Limit Theorem, Visually

And so on.



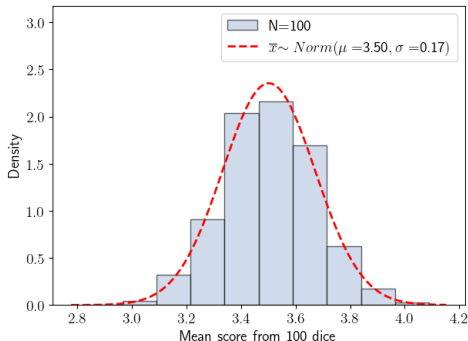
The Central Limit Theorem, Visually

As the number of RVs grows, we start to see the means (or sums) taking on a bell-shaped distribution. As $N \rightarrow \infty$ the distribution $\rightarrow \text{Norm}(\bar{x}, \sigma_{\text{sam}})$.



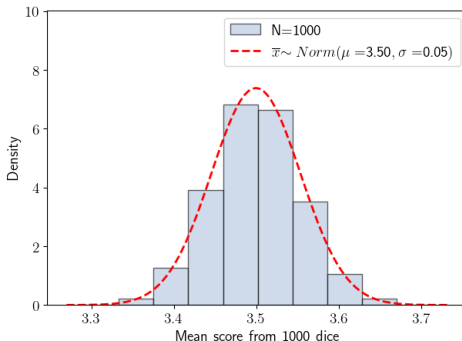
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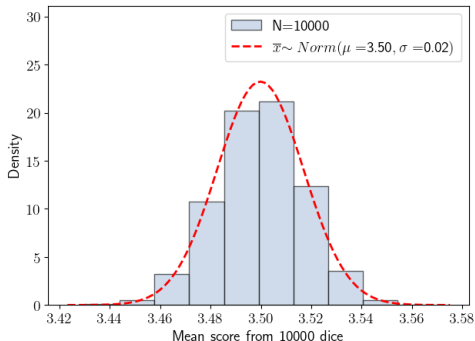
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The Central Limit Theorem, Visually

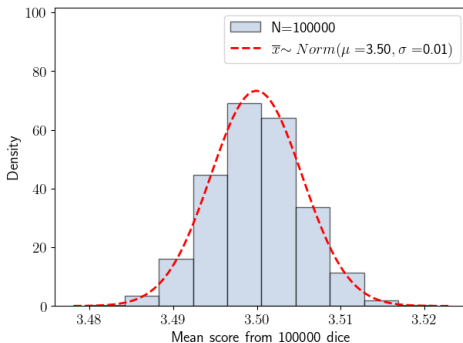
As the number of RVs grows, we start to see the means (or sums) taking on a bell-shaped distribution. As $N \rightarrow \infty$ the distribution $\rightarrow \text{Norm}(\bar{x}, \sigma_{\text{sampl}})$.



The Central Limit Theorem, Visually

The distribution of means will approach a normal distribution in the limit $N \rightarrow \infty$, but in practice the distribution starts looking Gaussian much earlier.

The **Central Limit** is the true mean $E[X] \equiv \mu$, which is quite stable as we increase N .



The CLT: Standard Normal RV

Recall the Standard Normal RV: $Z = \frac{X - \mu}{\sigma} \equiv \frac{X - E[X]}{\sqrt{V[X]}}$.

This is a particularly nice RV because $E[Z] = 0$ and $V[Z] = 1$ by definition, and it is a piece of cake to calculate probabilities with the Standard normal CDF (not analytically!)

```
# default params for scipy.stats norm are loc=0, scale=1
```

```
P(|Z| ≤ 1) = 68.27% norm.cdf(1)
```

```
P(|Z| ≤ 2) = 95.45% norm.cdf(2)
```

```
P(|Z| ≤ 3) = 99.73% norm.cdf(3)
```

CLT: we can use this cake-walk method to estimate probabilities for sums of means of RVs from any distribution.

The mean of N RVs as a Standard Normal RV

$$Z = \frac{X - \mu}{\sigma} \equiv \frac{X - E[X]}{\sqrt{V[X]}} \quad \text{We will let } X \rightarrow \bar{X}.$$

Denominator: $\sqrt{V[\bar{X}]} = \sqrt{V[X]/N} = \sigma/\sqrt{N}$

The mean of N RVs as a Standard Normal RV

$$Z = \frac{X - \mu}{\sigma} \equiv \frac{X - E[X]}{\sqrt{V[X]}} \quad \text{We will let } X \rightarrow \bar{X}.$$

$$\text{Denominator: } \sqrt{V[\bar{X}]} = \sqrt{V[X]/N} = \sigma/\sqrt{N}$$

$$\text{Numerator: } \bar{X} - E[\bar{X}] = \frac{1}{N} \sum_i^N X_i - E\left[\frac{1}{N} \sum_i^N X_i\right]$$

The mean of N RVs as a Standard Normal RV

$$Z = \frac{X - \mu}{\sigma} \equiv \frac{X - E[X]}{\sqrt{V[X]}} \quad \text{We will let } X \rightarrow \bar{X}.$$

$$\text{Denominator: } \sqrt{V[\bar{X}]} = \sqrt{V[X]/N} = \sigma/\sqrt{N}$$

$$\text{Numerator: } \bar{X} - E[\bar{X}] = \frac{1}{N} \sum_i^N X_i - E\left[\frac{1}{N} \sum_i^N X_i\right]$$

Letting $Y = \sum_i^N X_i$, and noting $N/\sqrt{N} = \sqrt{N}$, we have:

$$Z_N = \frac{Y - E[Y]}{\sqrt{N} \sigma}$$

Probabilities for sum or mean of RVs

$$Z_N = \frac{Y - E[Y]}{\sqrt{N}\sigma}$$

To calculate probabilities for our $Y = \sum_i^N X_i$, we use Z_N :

$$P(y_{low} \leq Y \leq y_{high}) = P\left(\frac{y_{low} - E[Y]}{\sqrt{N}\sigma} \leq Z_N \leq \frac{y_{high} - E[Y]}{\sqrt{N}\sigma}\right)$$

This looks a bit much, but is very easy to handle if we know N , $E[Y]$, and σ . See python example (just a few lines).

Any measurement can be thought of as a Random Variable X_i . The sum of IID* X_i is also a Random Variable Y . This will follow Norm(0,1) if the number of measurements of X_i used to construct it is large enough.

*Particle Physicists and other cowboys have been known to bend the rule that X_i must be IID with very pleasing results!

Hypotheses

A **Hypothesis** is an educated guess at the explanation for some occurrence. It must be based on real observations and it must be testable.

For example, if I collect some data (and I do not know the underlying PDF) I may observe that it looks like exponential decay when I plot it, but it has some lumps and bumps, so something more interesting could possibly be going on...

The **Null Hypothesis** H_0 is the educated guess that the most obvious (boring?) explanation for the data is the correct one. For the example above, the Null Hypothesis could be that the data is indeed $\text{Expon}(\lambda)$.

The **Alternative Hypothesis** H_1 is an alternative to H_0 . For the example above, the Alternative Hypothesis would be that the lumps and bumps are exciting resonances that will win your research group a nobel prize and pay for a new espresso machine.

Bias

My tangible excitement at the idea of an alternative hypothesis should raise a big red flag.

Anyone who has done Harvard's excellent [Unconscious Bias tests](#) will know that humans are hopeless at being unbiased, even when we aren't aware that we have a bias.

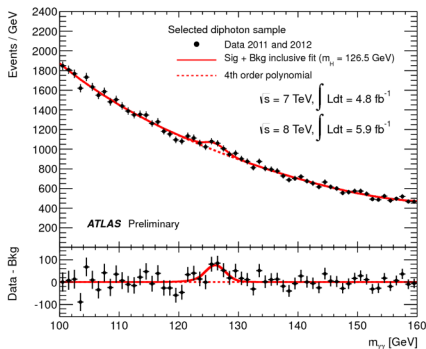
This means that if we are going to do excellent science, and not just come up with whatever result means we get a new espresso machine, we have to remove any chance of introducing bias.

One way of doing this is to set out very strict criteria for the null and alternate hypotheses before we look at our data.



Example of Null and Alternative Hypotheses

After waiting forever for the LHC to get going after some unfortunate stalls due to bad soldering and a ferret, we started to see 'Higgs-like' signs in 2011.



Strict Criteria[1]: Define a Critical Region

A **Critical Region** W is a region of the Sample Space in which the probability of finding the data, given the Null Hypothesis, is very small.

The region W can have any shape, making it suitable for tests involving multivariate data. The simplest possible definition of W for univariate data would be to draw a vertical line (a 'cut') and define W as being on one side of it.

A **Positive Result** means that we have found our data x in W , and so we **reject the Null Hypothesis**.

One cannot prove a hypothesis. Hypotheses can only be rejected or not rejected.

In reality it is impossible to find a W that is not consistent with both H_0 and H_1 . If we could, we would not need to do any of this....

Strict Criteria [2]: Significance Level

“Define a **Critical Region** W . This is a region of the Sample Space in which the probability of finding the data, given the Null Hypothesis, is **very small**.”

What does **very small** mean?

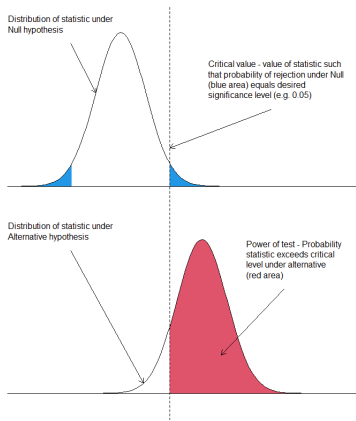
A popular choice in the wide world (but not in my world) is to set a tolerance level of 5%. We define our region W such that there is a 5% or lower probability of finding the data in that region, given the null hypothesis H_0 .

This is called the **Significance Level**, α .

$$P(x \in W | H_0) \leq \alpha$$

Once we when we have decided on W and α , we can look at (‘unblind’) our data.

Choosing a Critical Region is challenging



Credit: Fangz CC BY 4.0

Is purity or efficiency more important to you? High purity means small α (small contamination) and lower power (you have cut out a lot of your sample).

Are you suffering from a small dataset? If so, you will probably have to maximise power at the cost of purity.

Will your funding be withdrawn if you don't publish? This is not a valid reason to make a scientific decision, but is shockingly common (see later).

Limitations of the Statistical Test

If we find our data satisfies our predefined statistical test $P(x \in W|H_0) \leq \alpha$, then we have some responsibility to publish.

- With a Significance Level of $\alpha = 5\%$, this would give me major heeby jeebies.
- The statistical test for rejecting H_0 has a **False Positive** rate of up to 5%, which to me seems enormous. A one in twenty chance that this is...wrong.
- This is called a **Type 1 Error**.

If we find our data fails our predefined statistical test, then we have no reason to reject H_0 .

- This does not mean H_0 is correct; a hypothesis cannot be shown to be correct. It could be that we just got unlucky - the data ended up in some other critical region that we did not choose for our predefined statistical test.
- This is a **False Negative** or **Type 2 Error**

The Confusion Matrix

		Result of Statistical test	
		Positive $x \in W$	Negative $x \notin W$
Truth	H_0 is wrong	True Positive We have correctly rejected H_0	False Negative We should have rejected H_0 but got unlucky/ need more data.
	H_0 is correct	False Positive We have incorrectly rejected H_0	True Negative We have correctly kept H_0 on the table as a valid hypothesis

The Confusion Matrix

		Result of Statistical test	
		Positive $x \in W$	Negative $x \notin W$
Truth	H_0 is wrong	True Positive We have correctly rejected H_0	False Negative We should have rejected H_0 but got unlucky/ need more data.
	H_0 is correct	False Positive We have incorrectly rejected H_0	True Negative We have correctly kept H_0 on the table as a valid hypothesis

The Confusion Matrix

Probability of true positive: Power $P(x \in W H_0') = 1 - \beta$		Result of Statistical test	
		Positive $x \in W$	Negative $x \notin W$
Truth	H_0 is wrong	True Positive We have correctly rejected H_0	False Negative We should have rejected H_0 but got unlucky/ need more data.
	H_0 is correct	False Positive We have incorrectly rejected H_0	True Negative We have correctly kept H_0 on the table as a valid hypothesis

The Confusion Matrix **Type 2 Error**

		Result of Statistical test	
		Positive $x \in W$	Negative $x \notin W$
Truth	H_0 is wrong	True Positive We have correctly rejected H_0	False Negative We should have rejected H_0 but got unlucky/ need more data.
	H_0 is correct	False Positive We have incorrectly rejected H_0	True Negative We have correctly kept H_0 on the table as a valid hypothesis

The Confusion Matrix **Type 2 Error**

Result of Statistical test		Probability of false negative:	
		$P(x \notin W H'_0) = \beta$	
		Positive $x \in W$	Negative $x \notin W$
Truth	H_0 is wrong	True Positive We have correctly rejected H_0	False Negative We should have rejected H_0 but got unlucky/ need more data.
	H_0 is correct	False Positive We have incorrectly rejected H_0	True Negative We have correctly kept H_0 on the table as a valid hypothesis

The Confusion Matrix

		Result of Statistical test	
		Positive $x \in W$	Negative $x \notin W$
Truth	H_0 is wrong	True Positive We have correctly rejected H_0	False Negative We should have rejected H_0 but got unlucky/ need more data.
	H_0 is correct	False Positive We have incorrectly rejected H_0	True Negative We have correctly kept H_0 on the table as a valid hypothesis

The Confusion Matrix

Result of Statistical test		Probability of true negative: $P(x \notin W H_0) = 1 - \alpha$	
		Positive $x \in W$	Negative $x \notin W$
Truth	H_0 is wrong	True Positive We have correctly rejected H_0	False Negative We should have rejected H_0 but got unlucky/ need more data.
	H_0 is correct	False Positive We have incorrectly rejected H_0	True Negative We have correctly kept H_0 on the table as a valid hypothesis

The Confusion Matrix **Type 1 Error!**

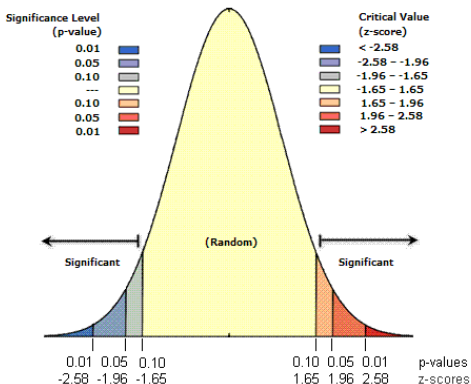
		Result of Statistical test	
		Positive $x \in W$	Negative $x \notin W$
Truth	H_0 is wrong	True Positive We have correctly rejected H_0	False Negative We should have rejected H_0 but got unlucky/ need more data.
	H_0 is correct	False Positive We have incorrectly rejected H_0	True Negative We have correctly kept H_0 on the table as a valid hypothesis

The Confusion Matrix **Type 1 Error!**

Result of Statistical test		Probability of false positive: Significance level $P(x \in W H_0) = \alpha$	
		Positive $x \in W$	Negative $x \notin W$
Truth	H_0 is wrong	True Positive We have correctly rejected H_0	False Negative We should have rejected H_0 but got unlucky/ need more data.
	H_0 is correct	False Positive We have incorrectly rejected H_0	True Negative We have correctly kept H_0 on the table as a valid hypothesis

p-values and z-values

The Significance of a measurement is expressed in terms of either a p-value or z-value, both of which are calculated from the data.



p-values and Z-values

We know the z-value already: $z = \frac{x - \mu}{\sigma}$. This tells us the deviation of our data x from the expected value μ in units of 'number standard deviations from the mean' or $n\sigma$.

For example there is a 68% probability of observing normally distributed data within the $\pm 1\sigma$ region of the standard normal distribution.

The Standard Normal PDF: $f_Z(x - \mu; \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{z^2}{2}\right]$

The Standard Normal CDF: $F_Z(x - \mu; \sigma) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z \exp\left[-\frac{z^2}{2}\right] dz$

p-values and z-values

The Standard Normal CDF: $F_Z(x - \mu; \sigma) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z \exp\left[-\frac{z^2}{2}\right] dz$

We can't solve this integral analytically, so we look up the probabilities (or use python).

$P(|Z| \leq 1) = 68.27\%$, z-value is 1σ , p-value is $(1 - 68.27\%) = 0.3173$

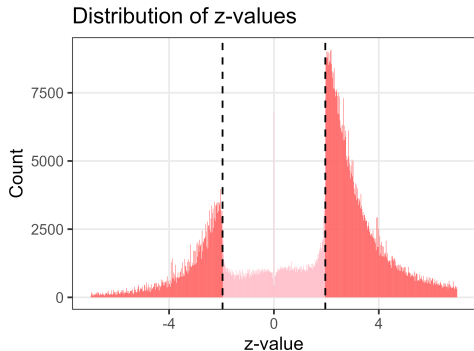
$P(|Z| \leq 2) = 95.45\%$, z-value is 2σ , p-value is $(1 - 95.45\%) = 0.0455$

$P(|Z| \leq 3) = 99.73\%$, z-value is 3σ , p-value is $(1 - 99.73\%) = 0.0027$

The p-value is the area under a portion of a tail of the standard normal distribution, where the normal distribution represents the null hypothesis. A very small p-value (high z-value) means it is very unlikely that we would find the data there, given the null hypothesis.

Poor Decision Making (medical publications)

This plot shows the distribution of z-values (significance levels) in published medical journals over the last fifty years.



Credit: Erik Van Zwet, Adrian Barnett

The data bear resemblance to a normal distribution (good) but most of the middle section (p-values greater than 5%) has been blown away.

This indicates **very bad science** (manipulating the tests/results) exacerbated by journals tending to reject work that fails to reject the null hypothesis, and funding relying on publications.

Confidence Levels, Limits, Intervals

- A **Confidence Level** often used is 95%, which corresponds to a p-value of 5% (the favourite of dodgy medical researchers) and a z-value of 1.644σ .
- The 95% **Confidence Interval** includes all values within 1.644σ of the mean of a normal distribution.
- The **Confidence Limits** in this case are $\mu + 1.644\sigma$ and $\mu - 1.644\sigma$

A CL of 95% means is that if we were to repeat the experiment 100 times, we would expect the true mean to lie in the Confidence Interval 95 times.

$$P(|Z| \leq 1.644) = 95\%, \text{ z-value is } 1.644\sigma, \text{ p-value is } (1 - 95\%) = 0.05$$

$$\text{p-value} = 1 - \text{norm.cdf}(\text{z-value}); \quad \text{z-value} = \text{norm.ppf}(1 - \text{p-value})$$

Thanks For Your Time

Fin.